

Experiments On 8051 Microcontroller Using EdSim51

Yogendra Pratap Singh (18/09/2026)

Question-1 Write a program to Add Two numbers 05H and 03H.

```
ORG 0000H

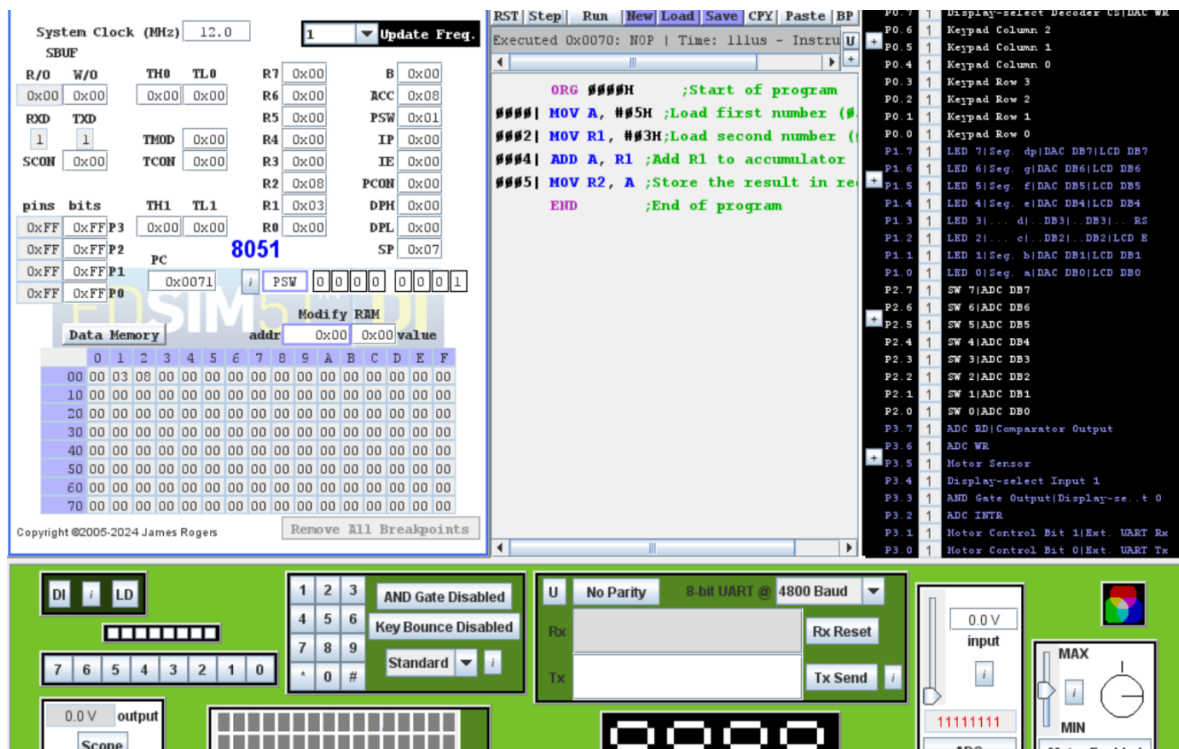
MOV A, #05H

MOV R1, #03H

ADD A, R1

MOV R2, A

END
```



Question-2 Write a program to Subtract Two numbers 05H and 07H.

```
ORG 0000H

0000| MOV A, #07H

0002| MOV R1, #05H

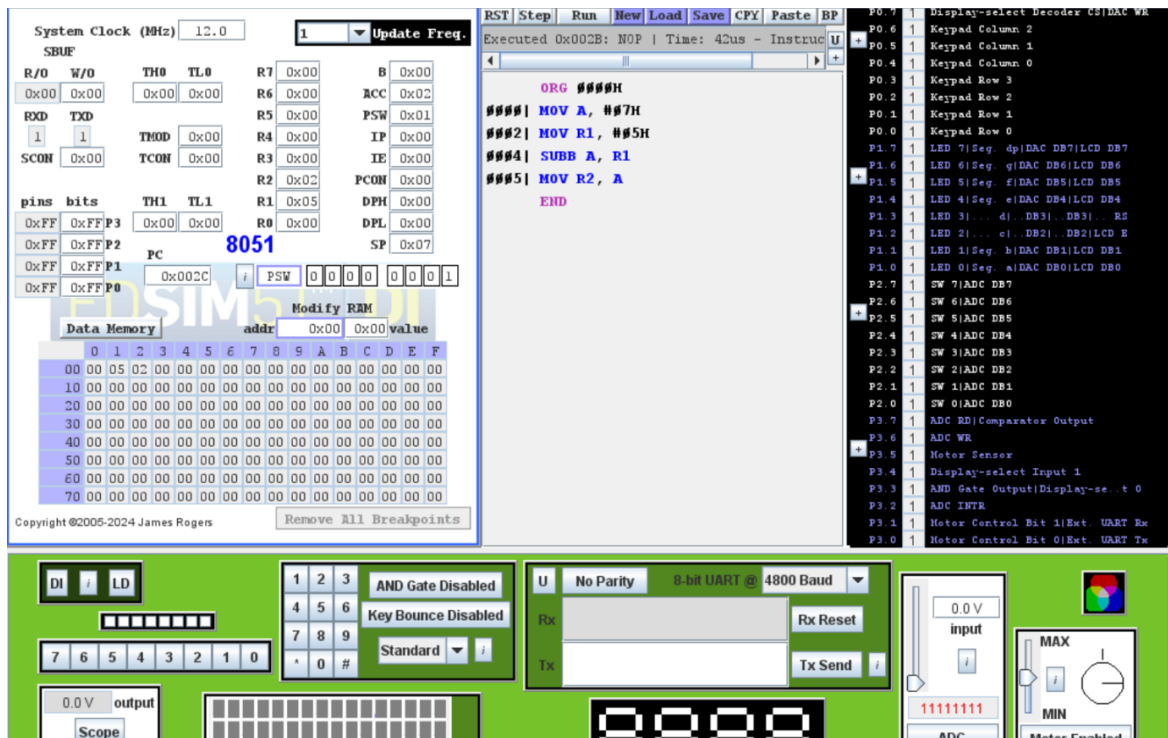
0004| SUBB A, R1

0005| MOV R2, A

END
```

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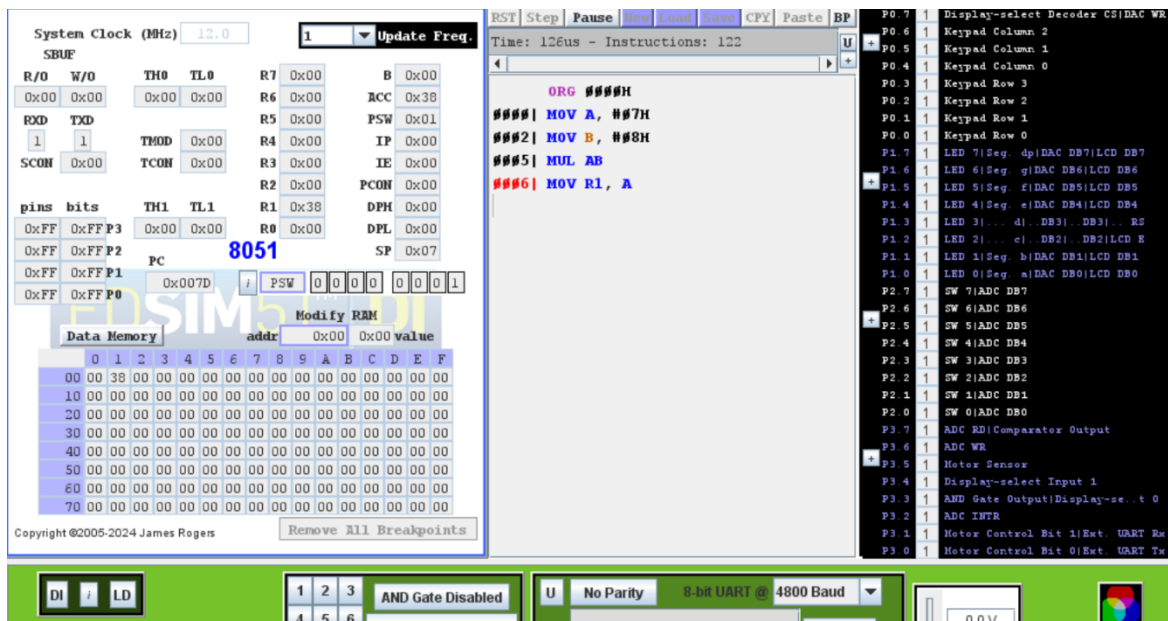
Question-3 Write a program to Multiply Two numbers 05H and 08H.

```

ORG 0000H

0000| MOV A, #07H
0002| MOV B, #08H
0005| MUL AB
0006| MOV R1, A

END
    
```

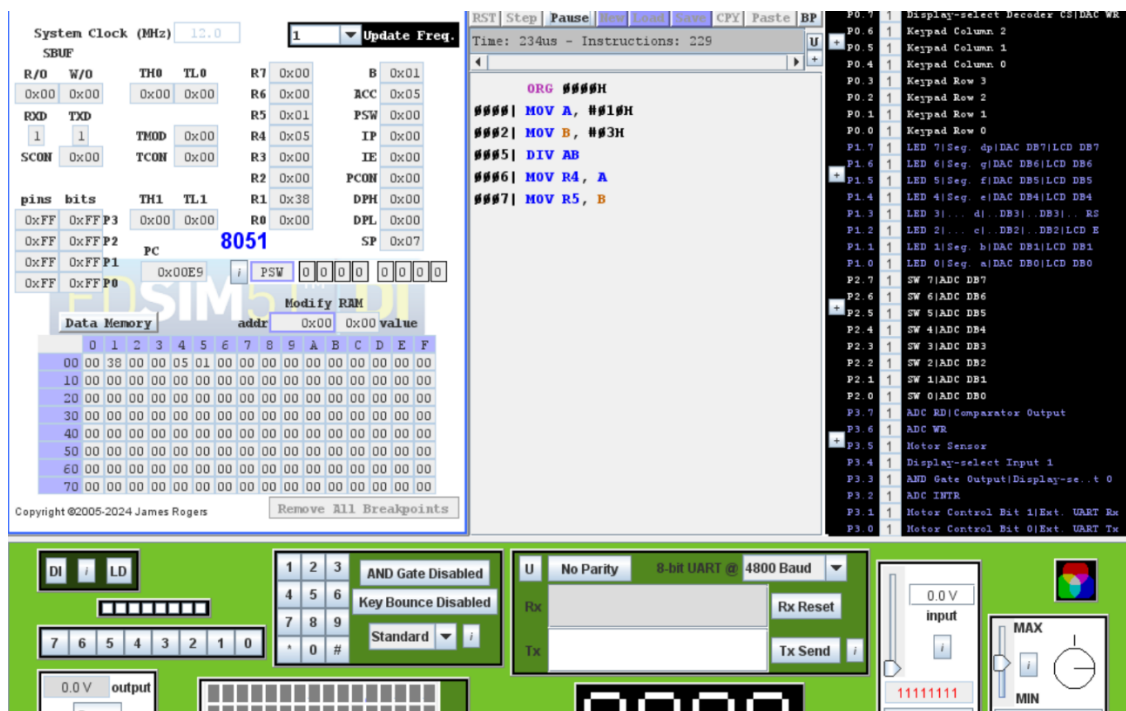


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Question-4 Write a program to divide 10H with 03H save the quotient in R4 and remainder in R5.

```
ORG 0000H  
0000| MOV A, #010H  
0002| MOV B, #03H  
0005| DIV AB  
0006| MOV R4, A  
0007 MOV R5, B
```



Question-5 Write a program to Swapping of Two numbers 55H is stored in R0 and AAH stored in R1 swap the content R0 and R1.

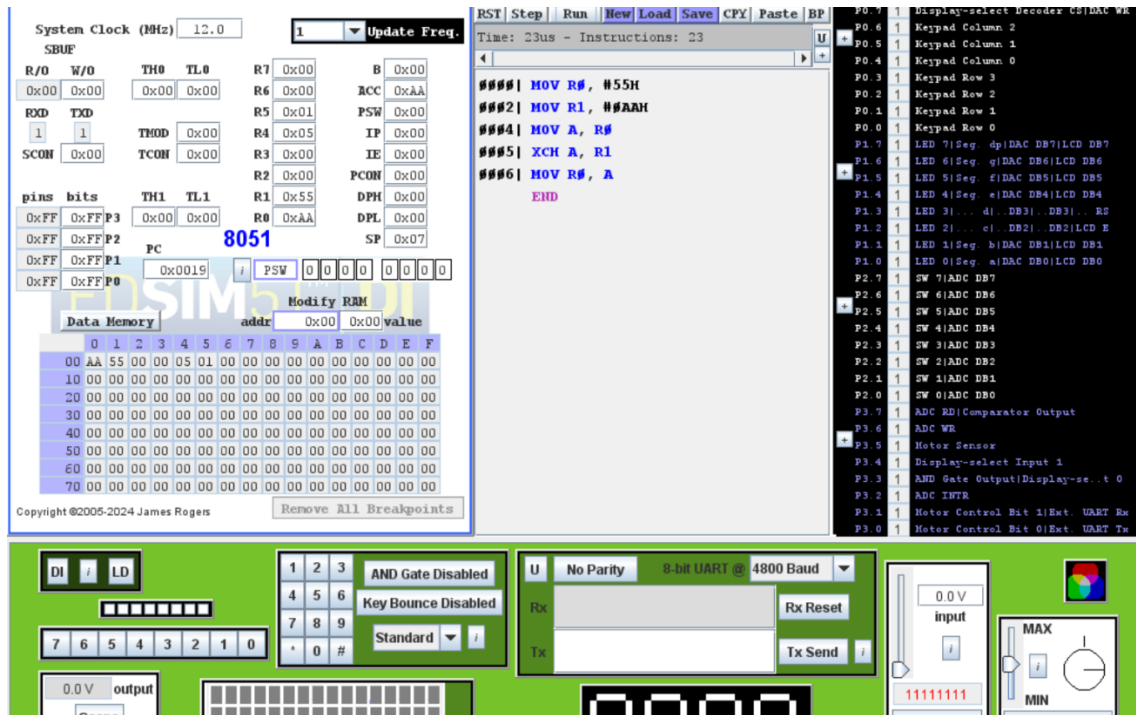
```
0000| MOV R0, #55H  
0002| MOV R1, #0AAH  
0004| MOV A, R0  
0005| XCH A, R1
```

Experiments On 8051 Microcontroller Using EdSim51

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0006| MOV R0, A

END



Question-6 Write a program to blink LED on Port 1.

ORG 0000H

0000| SETB P1.0

BLINK:

0002| CPL P1.0

0004| ACALL DELAY

0006| SJMP BLINK

DELAY:

0008| MOV R7, #0FFH

000A| MOV R6, #0FFH

000C| DJNZ R7, \$

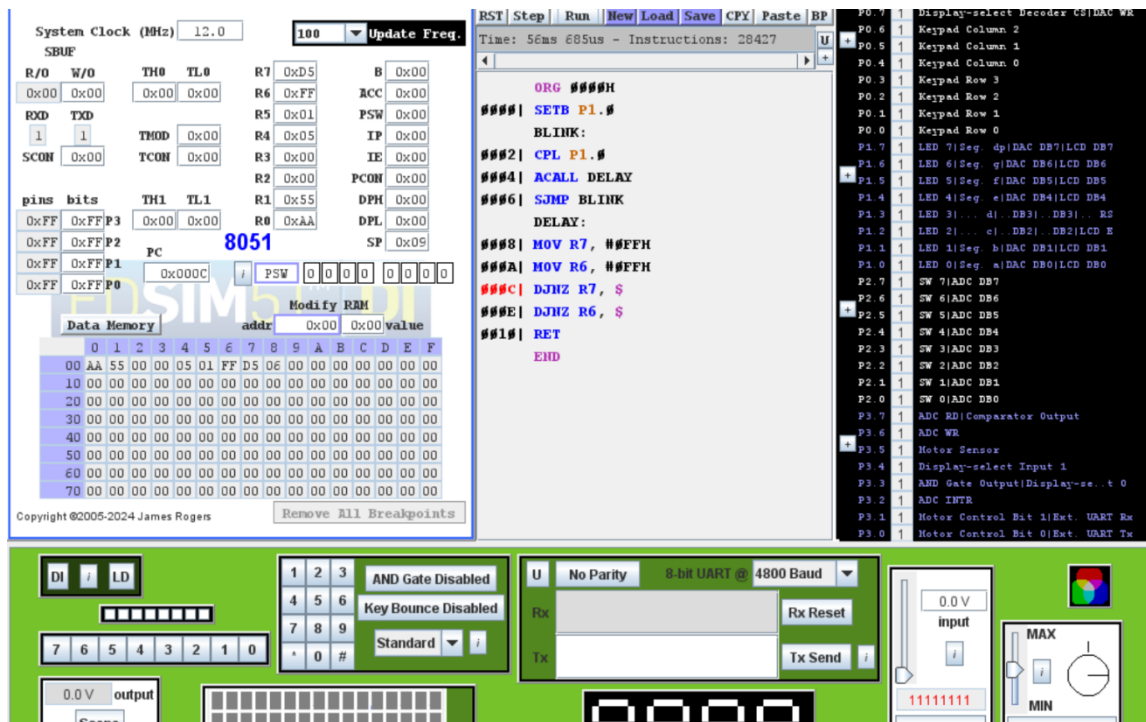
000E| DJNZ R6, \$

0010| RET

Experiments On 8051 Microcontroller Using EdSim51

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END



Question-7 Write a program to blink LED from right to left.

START:

0000| MOV A, #01H

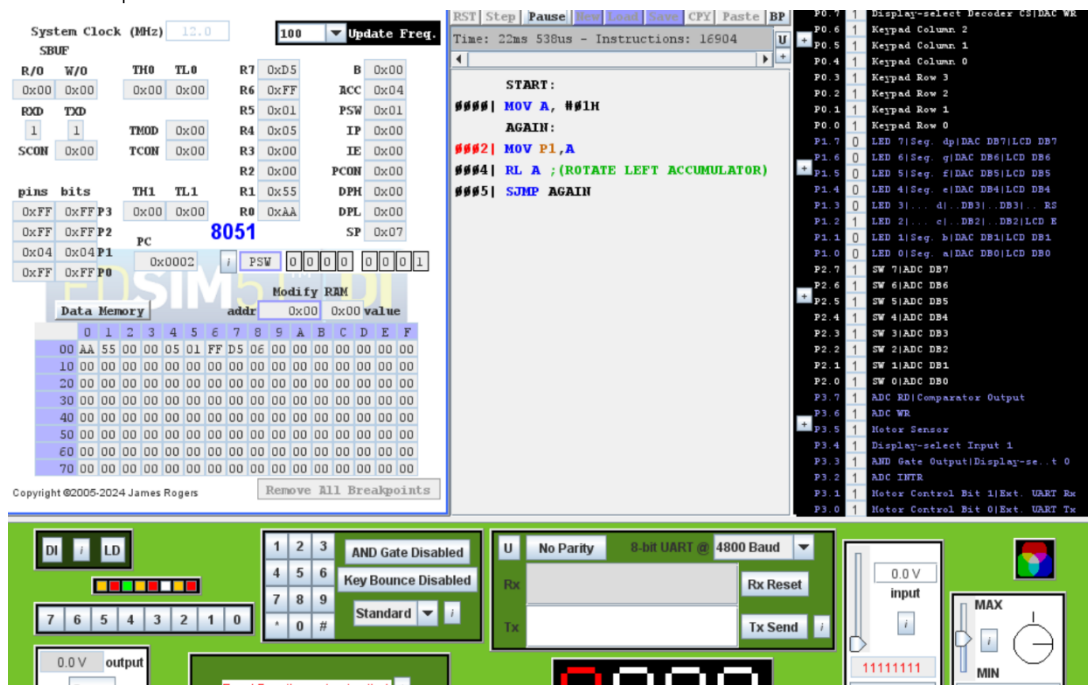
AGAIN:

0002| MOV P1, A

0004| RL A

0005| SJMP AGAIN

END



Experiments On 8051 Microcontroller Using EdSim51
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Question-8 Write a program to display 8051 on seven segment display.

```
ORG 0000H

START:

0000| SETB P3.3
0002| SETB P3.4
0004| MOV P1, #10000000B
0007| LCALL DELAY
000A| CLR P3.3
000C| MOV P1, #11000000B
000F| LCALL DELAY
0012| CLR P3.4
0014| SETB P3.3
0016| MOV P1, #10010010B
0019| LCALL DELAY
001C| CLR P3.3
001E| MOV P1, #11111001B
0021| LCALL DELAY
0024| JMP START

DELAY:

0026| MOV R0, #300
0028| DJNZ R0,$
002A| RET

END
```

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The screenshot displays the EdSim51 8051 microcontroller simulator interface, which is divided into several functional areas:

- System Configuration:** Located at the top left, it includes a 'System Clock (MHz)' set to 12.0, a frequency dropdown set to 10000, and an 'Update Freq.' button.
- Registers and Special Function Registers (SFRs):** A grid on the left shows the status of various registers (R0-R7, ACC, PSW, IP, IE, PCON, DPH, DPL, SP) and SFRs (TMO, TLO, TCON, TMOD, TH0, TL0, TH1, TL1, PC, PSW, P0-P3).
- Data Memory:** A table below the registers shows memory addresses from 00 to 70 and their corresponding hexadecimal values.
- Assembly Code Editor:** The central pane displays the assembly code being executed, starting with 'START:' and including instructions like 'SETB P3.3', 'MOV P1, #1', 'LCALL DELAY', and 'JMP START'. A 'DELAY:' section follows with 'MOV R0, #3' and 'DJNZ R0, \$'. The code ends with 'END'.
- IO and Hardware Status:** On the right, a list of I/O devices and their current states is shown, including 'Display-select Decoder CS/DAC WR', 'Keypad Column 2', 'Keypad Column 1', 'Keypad Column 0', 'Keypad Row 3', 'Keypad Row 2', 'Keypad Row 1', 'Keypad Row 0', 'LED 7(Seg. dp)/DAC DB7/LCD DB7', 'LED 6(Seg. g)/DAC DB6/LCD DB6', 'LED 5(Seg. f)/DAC DB5/LCD DB5', 'LED 4(Seg. e)/DAC DB4/LCD DB4', 'LED 3(Seg. d)/DAC DB3/LCD DB3', 'LED 2(Seg. c)/DAC DB2/LCD DB2', 'LED 1(Seg. b)/DAC DB1/LCD DB1', 'LED 0(Seg. a)/DAC DB0/LCD DB0', 'SW 7/ADC DB7', 'SW 6/ADC DB6', 'SW 5/ADC DB5', 'SW 4/ADC DB4', 'SW 3/ADC DB3', 'SW 2/ADC DB2', 'SW 1/ADC DB1', 'SW 0/ADC DB0', 'ADC ED/Comparator Output', 'ADC WR', 'Motor Sensor', 'Display-select Input 1', 'AND Gate Output/Display-select 0', 'ADC INTR', 'Motor Control Bit 1/Ext. UART Rx', and 'Motor Control Bit 0/Ext. UART Tx'.
- Hardware Components:** The bottom section features a green background with various hardware components: a 7-segment display showing 'LD', a keypad with 'AND Gate Disabled' and 'Key Bounce Disabled' status, a UART interface with 'No Parity' and '8-bit UART @ 4800 Baud' settings, a voltage meter showing '0.0 V', and a potentiometer with 'MAX' and 'MIN' settings.